

Demonstrating p5.fab: Direct Control of Digital Fabrication Machines from a Creative Coding Environment

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```
33 }
34
35 // function fabDraw() {
36 // printer setup
37 fab.setRelative();
38 fab.fanOff();
39 fab.moveTo(0,0);
40 fab.retract();
41 fab.setTemp(200, 50); // wait for bed to heat up
42 fab.introline(); // line back and forth to clean nozzle
43
44 // sculpture parameters
45 let startHeight = 0.2;
46 let a = 1.5;
47 let s = 1500;
48 let x = 90;
49 let y = 125;
50 let sf = 0;
51 let maxHeight = 100;
52 let l = 50;
53 let lineE = 4;
54 let dotE = 10;
55 a = 1;
56 fab.moveTo(x, y, startHeight); // move to start
57 for (let h = startHeight; h <= maxHeight; h += o) {
58 // line
59 fab.moveExtrude(x + 1 - sf, y, h, s, lineE);
60 fab.moveExtrude(x + 1 - sf, y + 1, h, s, lineE);
61 fab.moveExtrude(x + sf, y + 1, h, s, lineE);
62 fab.moveExtrude(x + sf, y, h, s, lineE);
63
64 // dots
65 fab.moveExtrude(x + sf, y, h + o, 25, dotE); // move slowly and extrude lots of files
66 fab.moveExtrude(x + 1 - sf, y, h + o, 25, dotE); // move quickly from point to point to rec
67 fab.moveExtrude(x + 1 - sf, y, h + o, 25, dotE);
68 fab.moveExtrude(x + 1 - sf, y + 1, h, 3 + s);
69 fab.moveExtrude(x + 1 - sf, y + 1, h + o, 25, dotE);
70 fab.moveExtrude(x + sf, y + 1, h, 3 + s);
71 fab.moveExtrude(x + sf, y + 1, h + o, 25, dotE);
72
73 fab.moveTo(x + sf, y, h + o + 0.3, 0); // pop up a bit to not collide
74 sf = a * sin(map(h, 0, maxHeight, 0, TWO_PI));
75 a += 0.2;
76
77 }
```

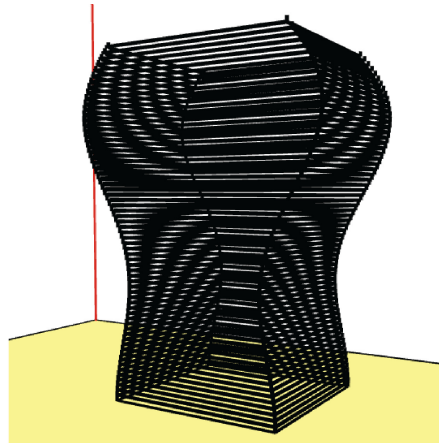


Figure 1: Our system, p5.fab, controls digital fabrication machines from the creative coding environment p5.js. We programmatically generate toolpaths using our library (left) which we can visualize in our user interface with a render command (center). We stream these commands to a desktop FFF 3D printer for fabrication; a photo of the resulting print is shown (right).

ABSTRACT

Machine settings and tuning are critical for digital fabrication outcomes. However, exploring these parameters is non-trivial. We seek to enable exploration of the full design space of digital fabrication. To do so, we built p5.fab, a system for controlling digital fabrication machines from the creative coding environment p5.js and informed by a qualitative study of 3D printer maintenance practices. p5.fab prioritizes material exploration, fine-tuned control, and iteration in fabrication workflows. We demonstrate p5.fab with examples of 3D prints that cannot be made with traditional 3D printing software, including delicate bridging structures and prints on top of existing objects.

CCS CONCEPTS

• Human-centered computing → Interactive systems and tools.

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KEYWORDS

Calibration, Interactive Fabrication, Computational Design, p5.js, 3D Printing, Maintenance, Craft

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1 INTRODUCTION

Computer-aided design and manufacturing software for using digital fabrication machines (such as CNC mills, laser cutters, and 3D printers) have steep learning curves and built-in assumptions about expected use [2]. HCI researchers have made a significant effort to decrease the barrier to entry through novel interaction techniques [e.g. 1, 4, 5]. However, contemporary software is designed for users to optimize toolpaths such that they will create a physical part that is as faithful to a model as possible. This assumption significantly limits the design space of digitally fabricated artifacts. Users cannot explore dynamic material behavior nor the effect of physical tools (e.g., end mills, nozzles, lenses), both of which impact outcomes as much as changing software settings. We identify an opportunity to better enable the exploration of digital form together with the real-world materiality of digital fabrication.

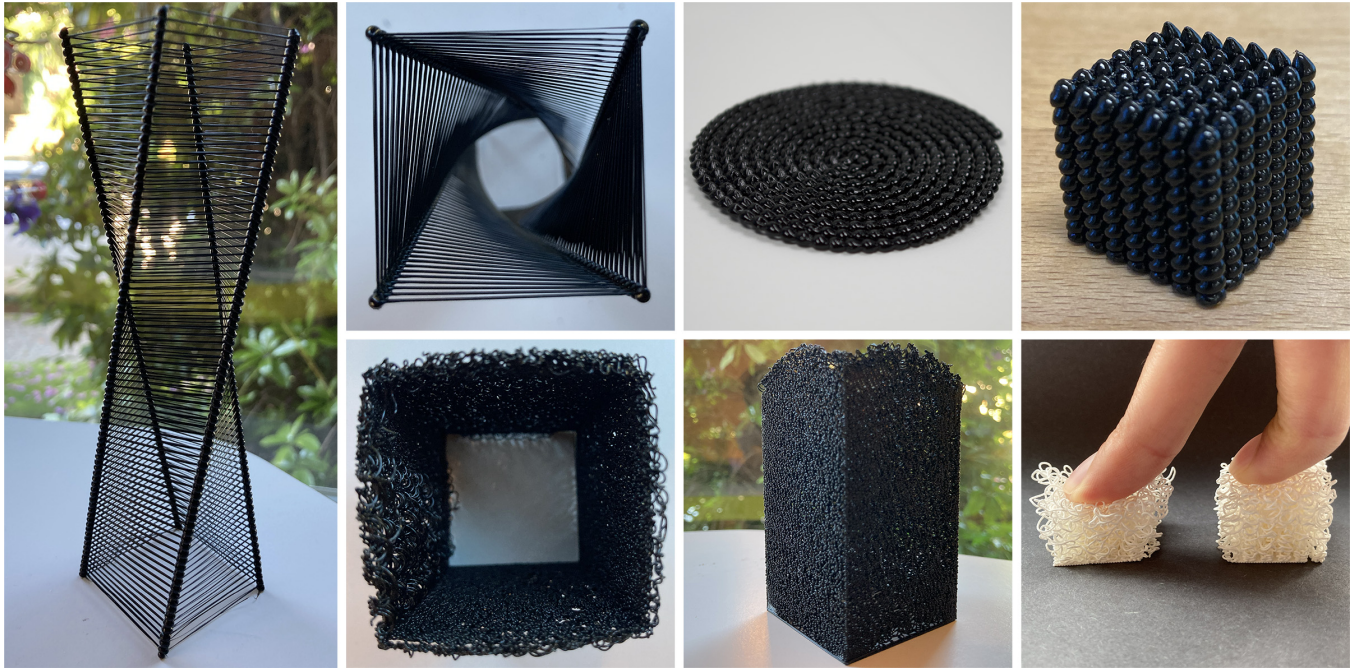


Figure 2: FFF 3D prints made using p5.fab. These prints would not be possible with off-the-shelf slicer software. Clockwise from left: bridged filament is anchored with one-shot extruded dots. Bridged filament top view. A 2D spiral of dots. A 3D dot print. Foam prints with varying density in compression. Foam print side view. Foam print top view.

Our paper demonstrates p5.fab, a system for controlling digital fabrication machines from the creative coding environment p5.js. p5.fab includes a library for programmatic control of machines within p5.js, a user interface to visualize toolpaths, and GUI elements for sending commands and receiving positions. Compared to the roundabout control offered by popular CAD and CAM software, we offer low-level control over machine toolpaths. These commands can be sent directly and immediately to the machine. The design goals of the system are informed by a formative study of 3D printer calibration and tuning practices [3].

We demonstrate our system through a series of 3D printing examples which center material exploration. These workflows include intricate structures built from single-width lines & over-extrusions and manual calibration techniques to print on top of existing objects in the work envelope. These structures would not be possible to fabricate using conventional 3D printing software. We cater to the craft sensibility fundamental to 3D printing practice to extend the space of what is possible to make with digital fabrication tools.

2 SYSTEM DESCRIPTION

Design Goals

Using insights from a formative study with 3D printing practitioners, we determined the following system design goals: (1) promoting practitioner agency through open-ended workflow exploration, (2) facilitating iterative material interaction, and (3) integrating digital design intent with physical making through toolpath control.

System Overview

Our system includes the p5.fab library and a user interface which includes a toolpath visualization and GUI elements for sending commands and receiving positions. Figure 3 shows the interface.

Direct Communication. We communicate with the printer from a computer using serial communication over a wired USB connection. We leverage WebSerial support which allows us to read and write serial data from the browser. While streaming, we can inject commands via the web console or custom user interface elements. These commands will be added to the top of the queue.

Toolpath Visualization. The system offers a simple visualization of the toolpaths. The system will parse the G-code generated by the operator and render to the screen using p5. The visualization is shown in reference to the machine’s work envelope, or the dimensional boundaries of the machine’s operating area. We monitor system state such as current position and temperature in real-time as data is received from the printer.

Toolpath Manipulation. In popular FFF 3D printers, an extruder can move within the work envelope by specifying XYZ coordinates. The extruder axis, which extrudes filament through the nozzle, can also be set to a position. Our system offers control over these motion settings: for example, moving to position (x, y, z) at speed s while moving the extrusion axis to position e : `moveExtrude(x, y, z, s, e)`; setting the max acceleration for each coordinate axis: `setMaxAcceleration(ax, ay, az, ae)`; and setting jerk for each coordinate axis: `setJerk(jx, jy, jz, je)`. We also expose easy access to a number of other convenient G-code commands (e.g. `autoHome`,

setNozzleTemp). Beyond built-in methods, we can send any arbitrary G-code to the machine. Some host applications offer a simple control panel to jog the printhead around the bed, send individual commands to the printer, and scaffold slightly more complicated behaviour such as pauses at specified layers. However, these features are for one-off use cases; by comparison, our system helps choreograph the large number of moves needed to print an entire model, access and procedurally manipulate higher order motion attributes, and consolidate bespoke multi-step workflows.

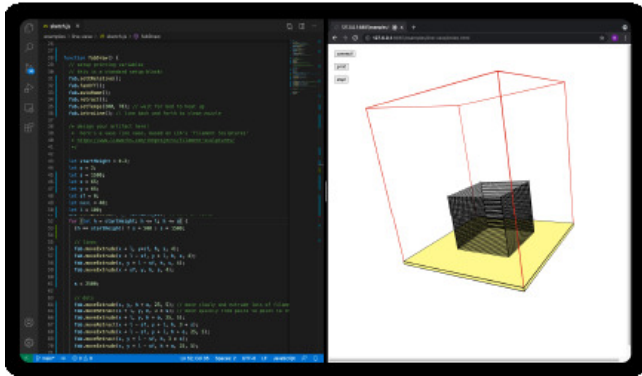


Figure 3: p5.fab interface. Toolpaths generated in code (left) are visualized in the user interface (right) then streamed to the machine over serial.

3 DEMONSTRATION WORKFLOWS

Our demonstration will showcase example workflows made possible with our system. These include intricate vases and foams shown in Figure 2, and a calibration routine to print on top of existing objects in the work envelope. Our demo will also allow visitors to craft custom toolpaths for immediate fabrication. Our system library is documented online and can be used with existing 3D printers using Marlin firmware with no printer modifications. This means visitors can continue their explorations of our system after leaving the conference.

Experimental Prints. Users can craft individual toolpaths which negotiate motion attributes (like velocity, acceleration, and jerk) with extrusion amount. These individual structures can be combined into more complex assemblies. For example, Figure 1 and Figure 2 (left) show vases which are created by first creating individual toolpaths to print single-width extrusions anchored by one-shot over-extruded dots. Similarly, sinusoidal toolpaths (Figure 2 bottom) create prints with textured surface finishes.

Unconventional Calibration. Users can use an interactive controls system to calibrate the printer to design and print directly onto existing objects, such as a handle onto a paper cup (Figure 4). This demo permits quick iteration of geometry shape and size without returning to CAD software.

4 CONCLUSION

We explored 3D printer calibration & tuning practices as a way to inform the design of interfaces for digital fabrication machines.

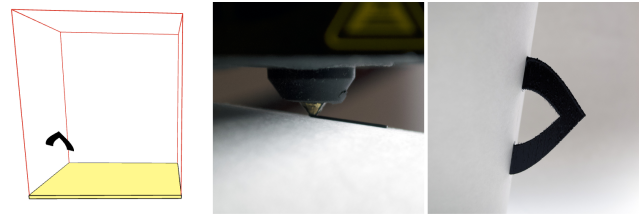


Figure 4: The p5.fab system makes it easy to print on top of existing objects, including ones that aren't flat. Shown here is a decorative handle being printed on a slanted cup. Left: after the operator aligns the nozzle to the desired points on the cup, the toolpath visualization shows the handle elevated from the print bed. Center: initial layer printing on the cup. Right: the resulting handle.

The material practice of calibration and tuning is supported by our contribution p5.fab, a system to control machines from the creative coding environment p5.js. We use our system to explore 3D printing in new ways, such as designing toolpaths with tuned motion attributes and printing onto existing surfaces for hybrid workflows. Given the size of the creative code community and increasing availability of personal fabrication machines, we believe p5.fab enables novel workflows which extend current opportunities in digital fabrication.

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